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SPACE SCIENCE WITH PYTHON

Space Science with Python — Bright Dots in the Dark Sky

Part 16 of the tutorial series describes another important basic concept in Space Science: the brightness of objects.

Photo by Unsplash

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Preface

This is the 16th part of my Python tutorial series "Space Science with Python". All codes that are shown here are uploaded on GitHub. Enjoy!

Take a look

Where do you live? In the city? In a rural town? In some remote place in the mountains?

And did you take a look at the night sky? There are vast differences of the night sky's appearance depending on where you live. Thanks to our modern lifestyle, the so-called light pollution erased the night's tapestry: The Stars.

Take a look outside this night. You see differently sized and coloured stars. You can try to classify the brightness of stars by comparing them with their neighbouring star. *Which one is brighter, or fainter?*

A 2000 year old definition ...

The old Greek did probably the same. And since stellar constellations were important for agriculture, nautical navigation or even astrology, they also started to classify the brightness of the stars. The idea: the brightest stars are Magnitude 1 stars and the faintest that are barely visible to the naked eye belong to the Magnitude 6. Further, a difference of 1 magnitude (short: *mag* or ^{m}) corresponds to a factor of 2 for the difference in brightness.

... And a modern re-definition

Modern astronomy is not limited to the unaided eye and the definition above cannot be applied mathematically since it highly depends on the subjective perception. The human senses or perceptions are mostly logarithmically and follow the Weber–Fechner law. A star with a magnitude 1 appears to be twice as bright as a star with a magnitude of 2. However, the corresponding *actual* flux density (power per area: W/m^2) does not scale linearly.

The modern magnitude definition of an object (obj) is shown below and is always compared to a reference (ref) source. *m* represents the magnitude and has no dimensionality. *I* is the corresponding flux density.

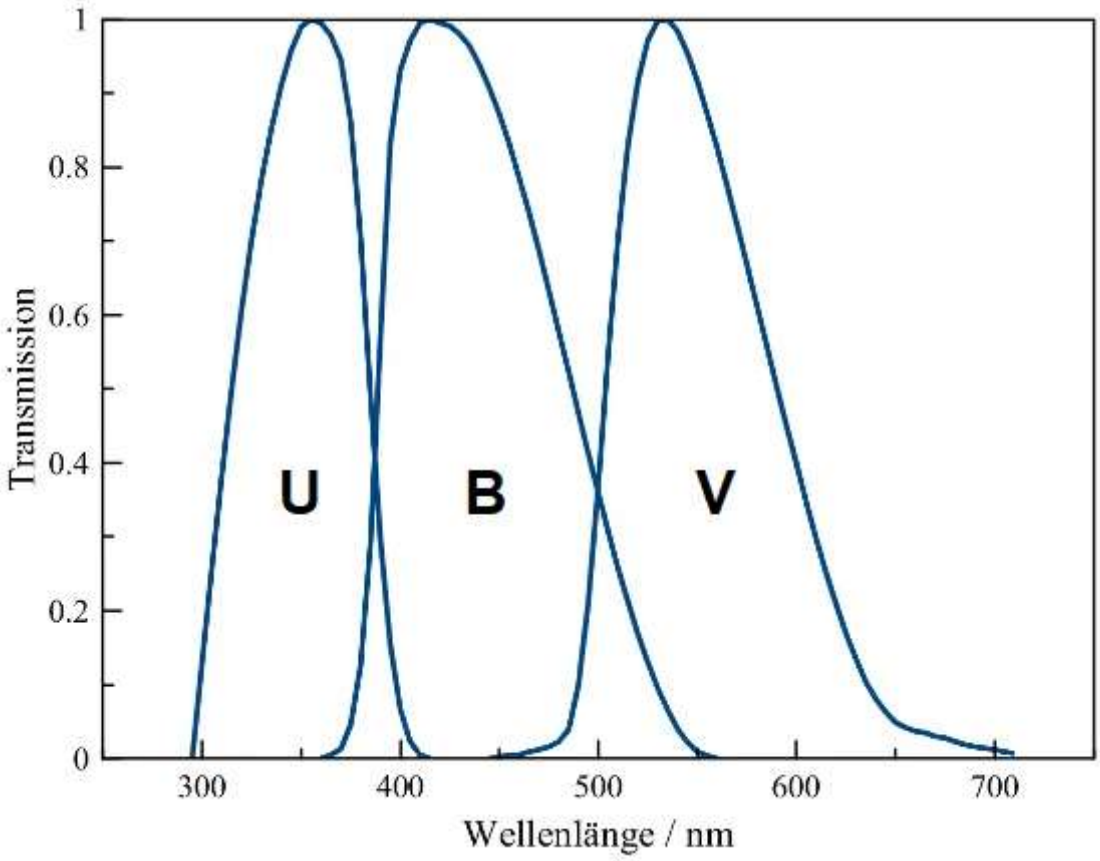
$$m_{\text{obj}} - m_{\text{ref}} = -2.5 \log_{10} \left(\frac{I_{\text{obj}}}{I_{\text{ref}}} \right)$$

Equation 1

Brightness computations can be very complex and depend on the scientific questions. The calculations may consider the solid angle, wavelength or different filters.

The following figure shows the standard UBV photometric system that shows only a fraction of all commonly used filters. The shown filter functions define a filter transmission depending on the light's wavelength:

- **U:** The ultra-violet filter has its peak at 364 nm
- **B:** The blue filter has its peak at 442 nm
- **V:** The yellow filter, or Visual filter has its peak at 540 nm and can be seen as a rough approximation of the human perception.



UBV photometric system. The transmission of the filters are plotted vs. the light's wavelength. Credit: Michael Oestreicher (Wikipedia); License: Creative Commons Attribution-Share Alike 4.0 International

Objects have different density fluxes depending on the filter. Simple filter relations like the value of **B** minus **V** indicate the colour of an object. A negative **B-V** value corresponds to a *bluish* and a positive value to a *reddish* colour. Based on these colour indices scientists can derive essential properties of stars or asteroids.

What is a good reference density flux? Is there an actual standard? Historically, the star Vega has been defined as the m = 0 reference star. However, the flux density varies for different filters and due to its natural variation, Vega's apparent magnitude in the V-Band filter is currently at around 0.03.

Vega (part of Lyra) is located between Cygnus and Hercules and currently be seen in the night sky. Screenshot from stellarium.

Anyway, let's set the reference magnitude to 0:

$$m_{\text{obj}} = -2.5 \log_{10} \left(\frac{I_{\text{obj}}}{I_0} \right)$$

Equation 2

Thankfully, the reference density flux for 0 mag was defined by the International Astronomical Union in the year 2015. On page 2 it is written that:

> *[…] to define the zero point of the apparent bolometric magnitude scale by specifying that m=0 mag corresponds to an irradiance or heat flux density of*

$$I_0 = 2.518021002 \cdot 10^{-8} \ \frac{\text{W}}{\text{m}^2}$$

Equation 3

So, this value corresponds to the flux density of Vega in the V Band, as seen from Earth. But do you have a *feeling* for these small values? 10^{-8} W/m^2 appears to be … well … let's compare it with something one can compare with: Our Sun. The flux

density of the Sun is (depending on the latitude and time) a few hundred Watt per square-meter. *WOW!* This is 11 orders of magnitude (!) larger than Vega's flux density. But we see Vega as one of the brightest stars at the night sky. The human eye and its perception performance are amazing.

Simplifying things

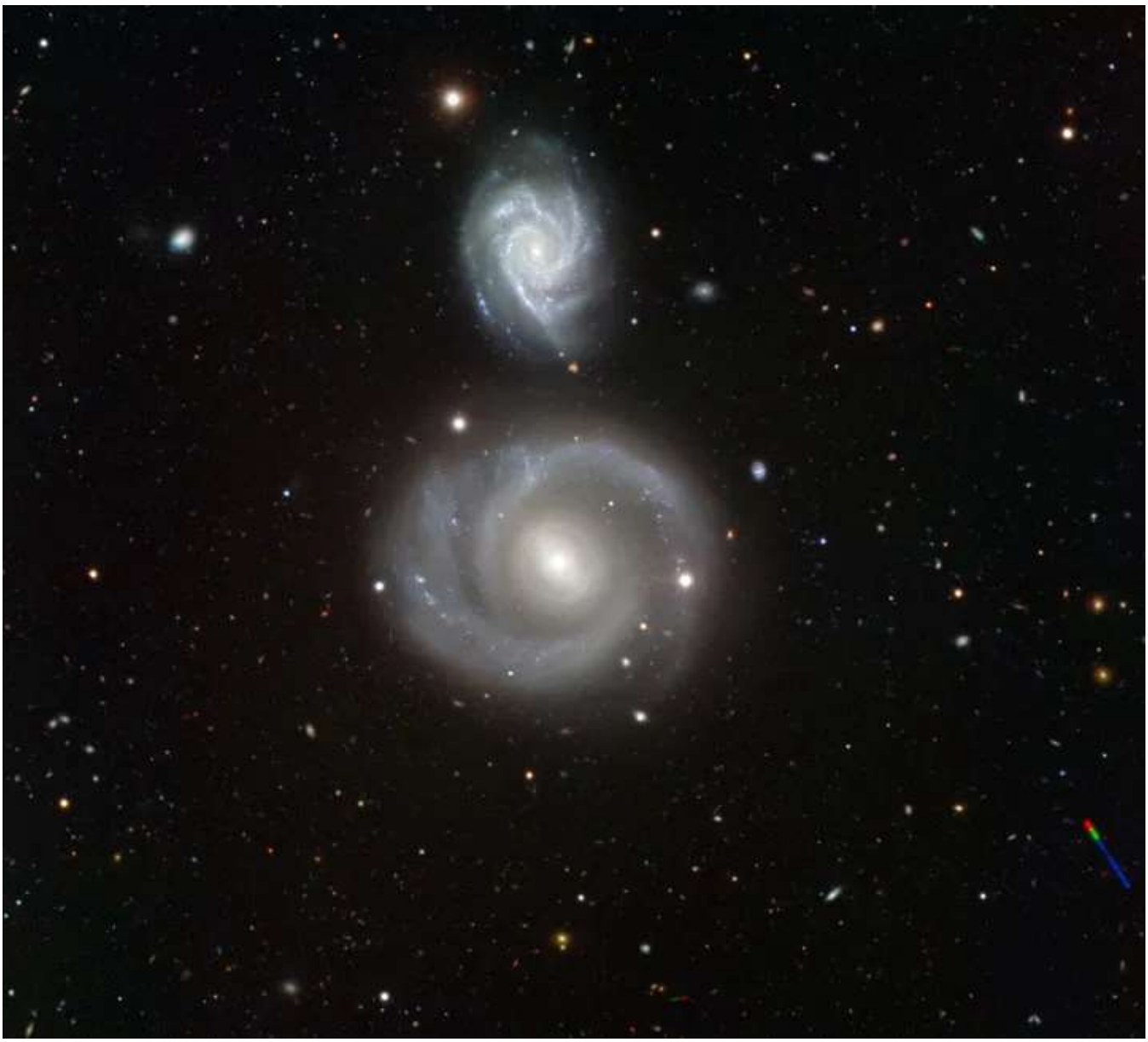
Let's insert the value from equation 3 to equation 2 and solve the equation to the flux density of the object (obj). We can approximate our solution to:

$$I_{\mathrm{obj}} \approx 10^{0.4(-m_{\mathrm{obj}}-19.0)}$$

Equation 4

Now we can easily insert a magnitude value (for the V-Band filter) in the equation and we get immediately the flux density in W/m^2. If you want to consider the atmospheric attenuation (that actually depends on the the azimuth angle, but let's keep it simple) you can replace 19.0 with 18.6.

Another flux density parameter is the surface brightness that is given in magnitudes per square-arcsecond. It is commonly used for *non-dot-like* objects like galaxies or comets. We will introduce and work with it in the future. In the meantime, and also for our future tutorial sessions, we will stay with the equation shown above. We will compute the brightness and visibility of asteroids or meteoroids to determine their detectability. Please consider also that astronomers compute the flux also depending on the wavelength! However, we will work with our simplification for the V-Band.



Do not appear a “dots” in the night sky: Galaxies. The brightness of these objects is mostly given in magnitudes per square-arcsecond. The galaxy on top is NGC 800, below is NGC 799. Credit: ESO; License: Creative Commons Attribution 4.0 International License

Some hands-on

Today, we will not aim for a specific task. This basic tutorial shall give us some feeling and understanding of this basic brightness concept. Let’s compute and visualise some values and create a lookup-table.

For this session, we only need *pandas* and *numpy* and create an empty dataframe that will store the lookup data (line 3).

Part 1/6

Let's create some functions. Since we will use the brightness computation in future tutorials, functions can be easily re-used and stored in some additional scripts. Coding part 2/6 is a function that converts the magnitude to the flux density respectively the irradiance in W/m^2. The optional input parameter *use_attn* triggers (if requested) an if-else clause in line 23 to 27 to add a atmospheric attenuation factor. Equation 4 is given in line 31 and returned afterwards.

Part 2/6

The irradiance can be easily converted to power by applying an area given in m^2. Such an area may represent the mirror size of the telescope. The power is computed

in line 19.

Part 3/6

Finally, power times time provides us the energy given in Joule (line 21). The time may represent the exposure time of a camera (like a Charge-Couple Device: CCD).

Part 4/6

Let's compute an exemplarily lookup-table for a magnitude range of -2 to +7 (range of the visible stars for the human eye, defined in line 2). The magnitudes are added to a column in line 5. Line 8 to 9 and 12 to 13 compute the magnitude corresponding irradiance without and with the atmospheric attenuation. Finally, we print the lookup-table. Let's take a look.

Part 5/6

The range of around 8 mag covers 3 orders of magnitude for the irradiance. The applied attenuation factor reduces the flux density by around 40 %.

	magnitude	irradiance [W/m^2]	irradiance attn [W/m^2]
0	-2.0	1.584893e-07	2.290868e-07
1	-1.0	6.309573e-08	9.120108e-08
2	0.0	2.511886e-08	3.630781e-08
3	1.0	1.000000e-08	1.445440e-08
4	2.0	3.981072e-09	5.754399e-09
5	3.0	1.584893e-09	2.290868e-09
6	4.0	6.309573e-10	9.120108e-10
7	5.0	2.511886e-10	3.630781e-10
8	6.0	1.000000e-10	1.445440e-10

Lookup-tables are sometimes helpful tools for night observation sessions where no computer or tablet are present. Feel free to add further parameters to create a more detailed and sophisticated list (e.g., if you have a telescope you can consider the collection area of the mirror, or consider certain exposure times).

Part 6/6

Conclusion & Outlook

The session today was a basic tutorial about the brightness computation in astronomy and space science. The logarithmic definition (Equation 1) may appear confusing, but can be simplified as shown in Equation 4.

In the daily science work, astronomers use Equation 1 to e.g., determine the magnitude of an unknown asteroid. They use huge star catalogues that list the magnitudes of several stars in different filters. By comparing the brightness of the unknown asteroid with the brightness of a close appearing star one can easily determine the asteroid's magnitude.

In future tutorial sessions, we will use our functions to compute brightness values of asteroids, comets and meteors. We will also model the complete *detection chain* of asteroids (from the Sunlight that reflects on the surface of the asteroid, to the 8-bit image that is being generated by a camera that is attached on a telescope).

Don't forget: On 30th June is Asteroid Day! We will start there with an asteroid and Near-Earth Objects session!

Thomas

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